

Do **not** write solutions on this page.

12. [Maximum mark: 21]

Consider the differential equation $\frac{dy}{dx} - y \operatorname{cosec} 2x = \sqrt{\tan x}$, where $0 < x < \frac{\pi}{2}$ and $y = \frac{\pi}{4}$ at $x = \frac{\pi}{4}$.

- (a) Use Euler's method with step length $\frac{\pi}{12}$ to find an approximate value of y when $x = \frac{5\pi}{12}$.

Give your answer correct to three significant figures. [3]

- (b) Show that $\frac{d}{dx} \left(\frac{1}{2} \ln(\cot x) \right) = -\operatorname{cosec} 2x$. [4]

- (c) Show that $\sqrt{\cot x}$ is an integrating factor for this differential equation. [4]

- (d) Hence, by solving the differential equation, show that $y = x\sqrt{\tan x}$. [5]

- (e) Consider the curve $y = x\sqrt{\tan x}$ for $0 < x < \frac{\pi}{2}$ and the Euler's method approximation calculated in part (a).

- (i) Find the y -coordinate at $x = \frac{5\pi}{12}$. Give your answer correct to three significant figures.

- (ii) By considering the gradient of the curve, suggest a reason why Euler's method does not give a good approximation for the y -coordinate at $x = \frac{5\pi}{12}$.

- (iii) State why this approximation is less than the y -coordinate at $x = \frac{5\pi}{12}$. [3]

- (f) By considering $\frac{dy}{dx} - y \operatorname{cosec} 2x = \sqrt{\tan x}$, deduce that the curve $y = x\sqrt{\tan x}$ has a positive gradient for $0 < x < \frac{\pi}{2}$. [2]

